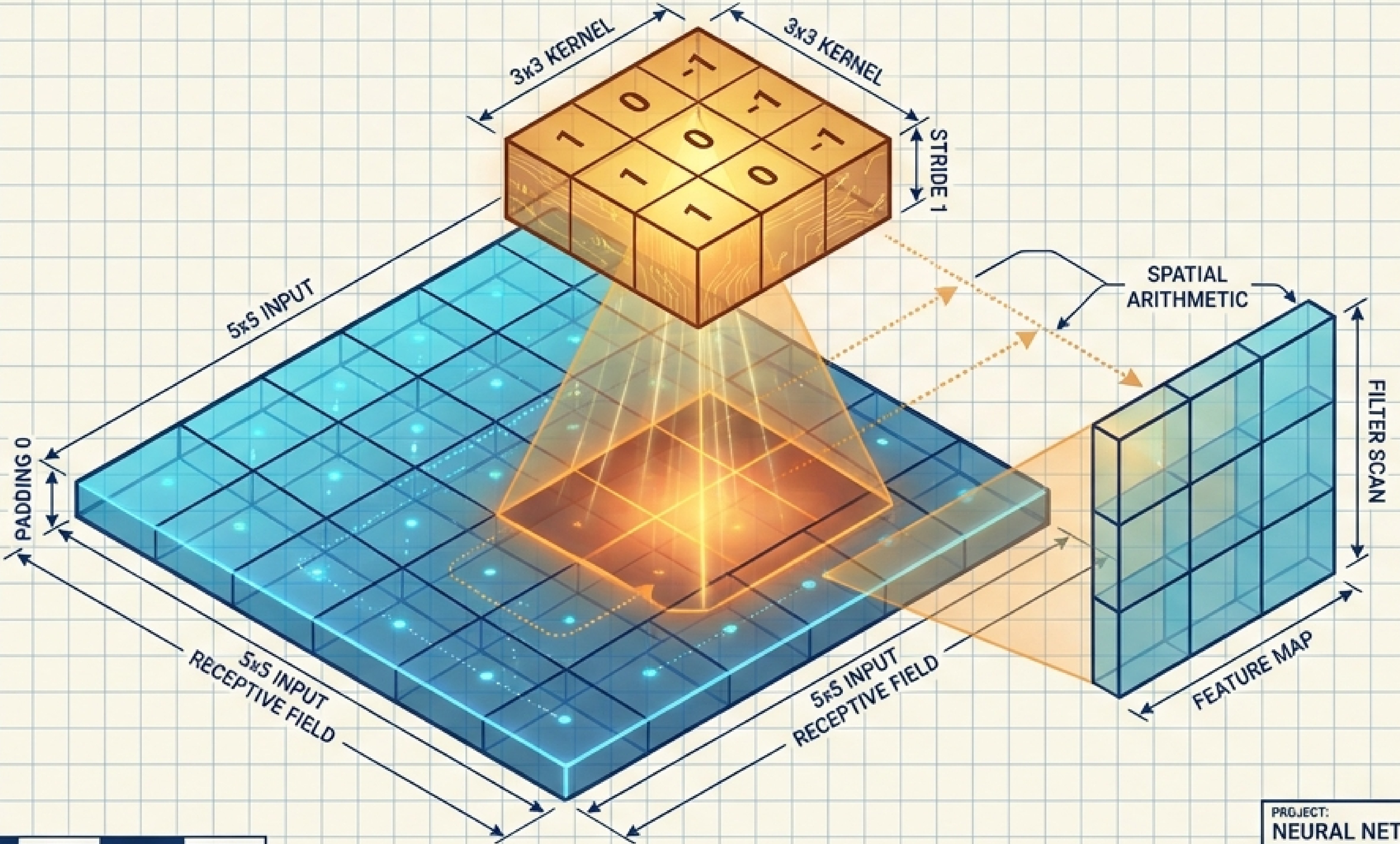
The Mechanics of Artificial Perception

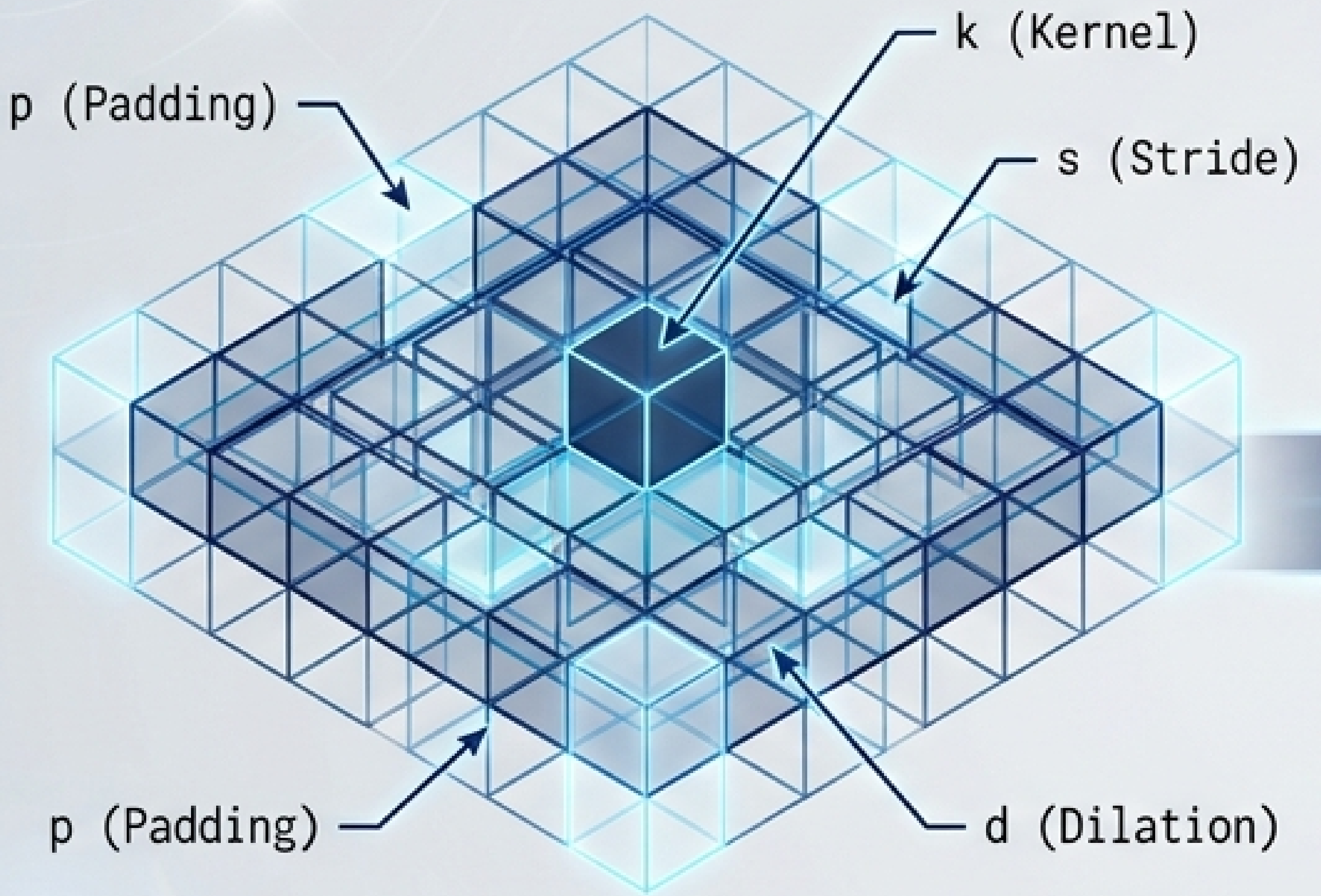
A visual guide to convolution, spatial arithmetic, and receptive fields in neural networks.



PROJECT: NEURAL NETWORKS ARCHITECTURE		
DRAWN BY: A.I. DESIGN SYSTEMS	DATE: OCT 26, 2024	SHEET NO: A-001 REV B

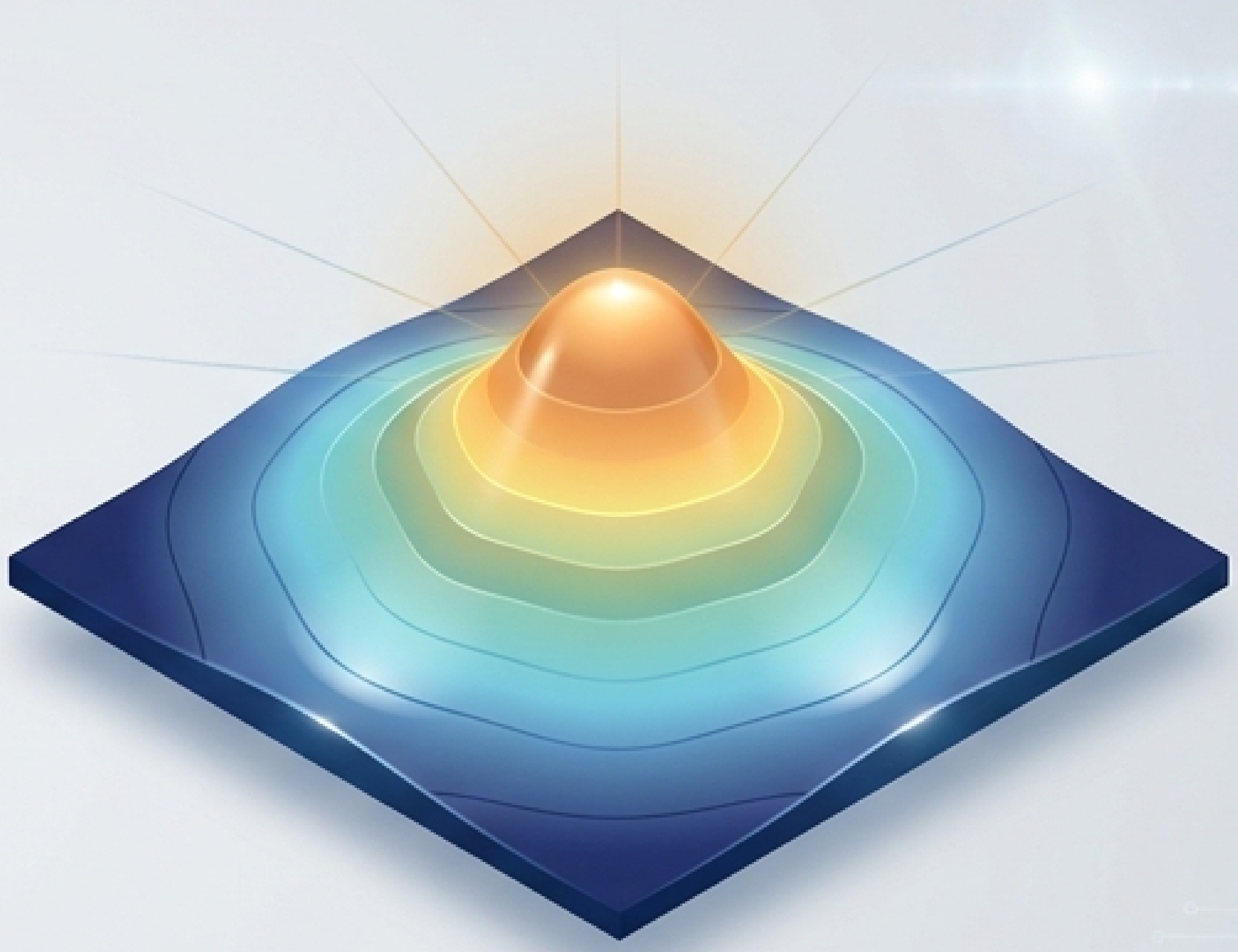
Building the Network's Eye From the Bottom Up

The Micro: Spatial Arithmetic



Master the pixel-level spatial operations that manipulate input matrices.

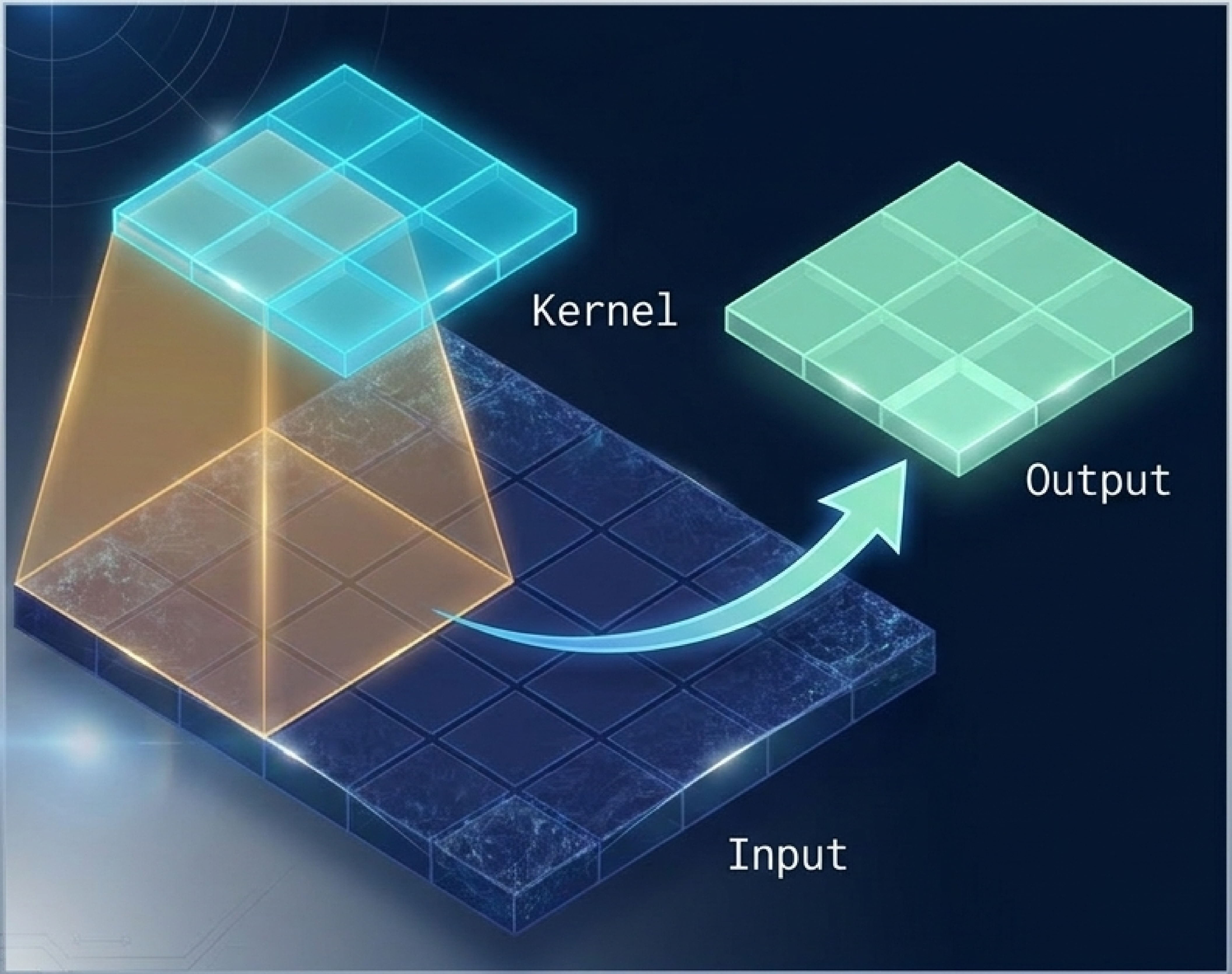
The Macro: Systemic Perception



Understand how these micro-operations compound to form the network's Effective Receptive Field.

A convolutional neural network is essentially a highly structured, grid-based optical system.
To understand its theoretical perception, we must first master its spatial engineering.

The Raw Glass: Understanding the Kernel (k)



The Operation

A discrete convolution is a local linear transformation. The kernel slides across the input, computing the dot product at each position.

The Problem

Without intervention, every convolution shrinks the feature map. A 3x3 kernel traversing a 5x5 input naturally produces a 3x3 output, because the kernel cannot slide beyond the boundaries.

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k = the kernel size along a
given axis.
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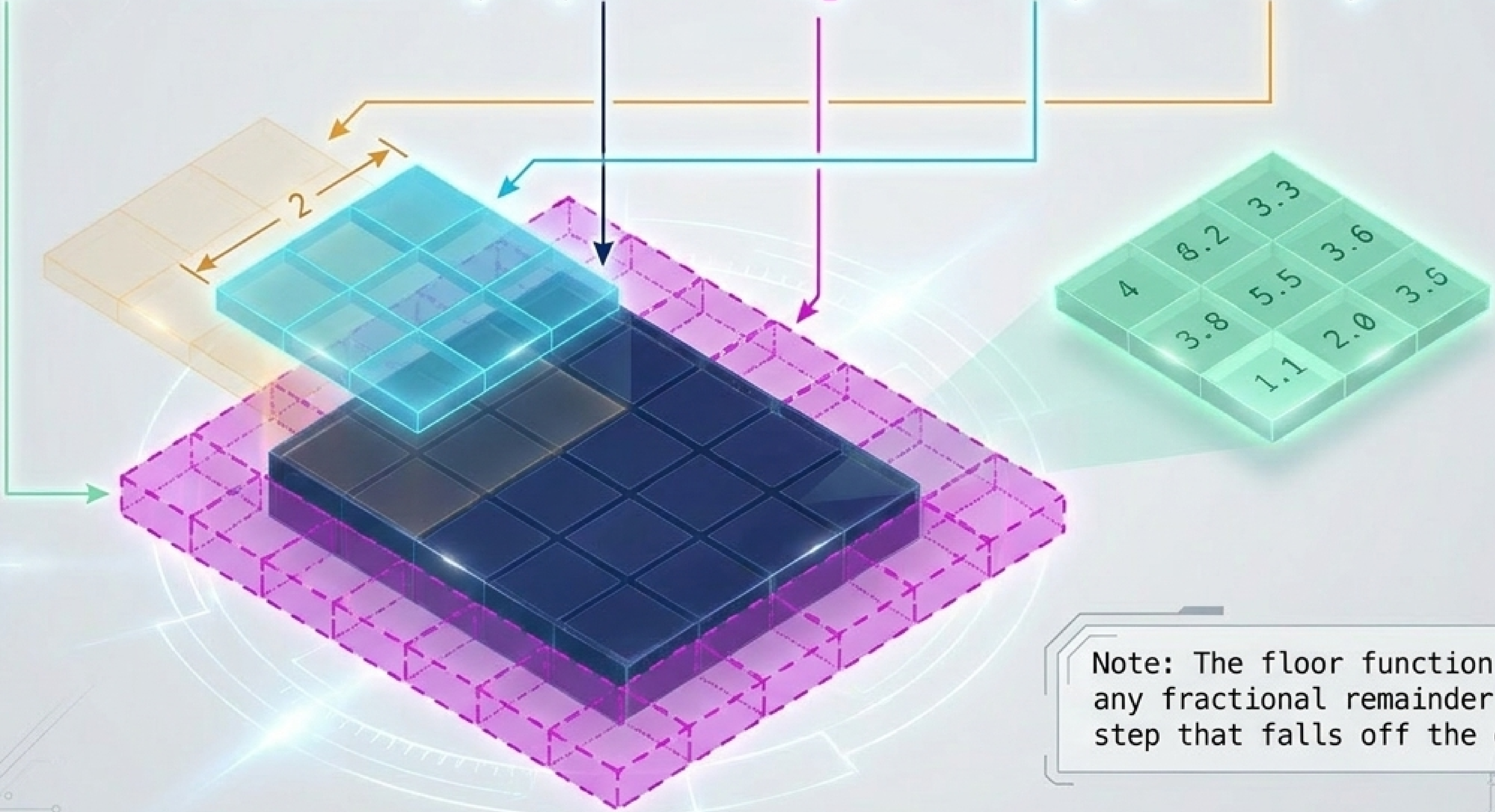
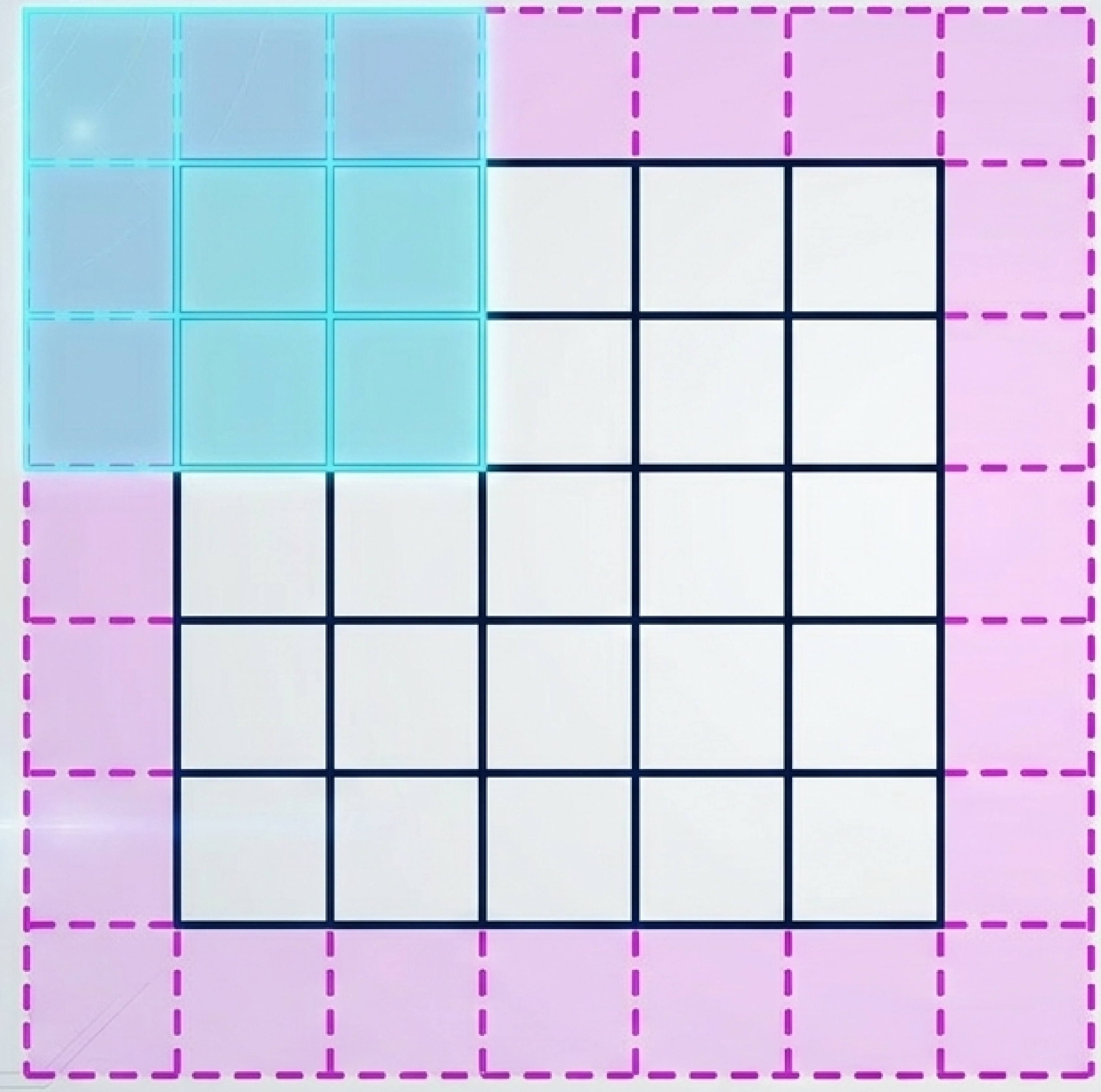
The Master Equation of Spatial Output

$$o = \text{floor}(\, (i + 2p - k) \,/\, s \,) + 1$$



Note: The floor function discards any fractional remainder of a stride step that falls off the grid.

The Phantom Border: Preserving Resolution with Padding (p)



The Edge Penalty

In standard convolution, corner pixels are only sampled once, while central pixels are sampled multiple times. The image shrinks and edge data is lost.

The Solution

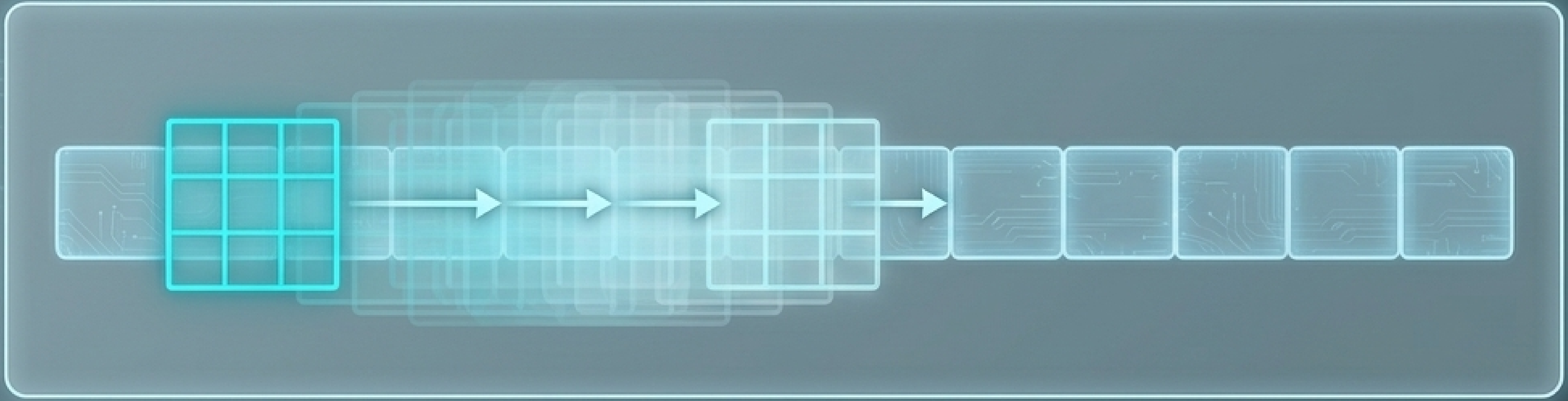
Padding the input with zeros artificially expands the effective input size to `i + 2p`.

Same Padding

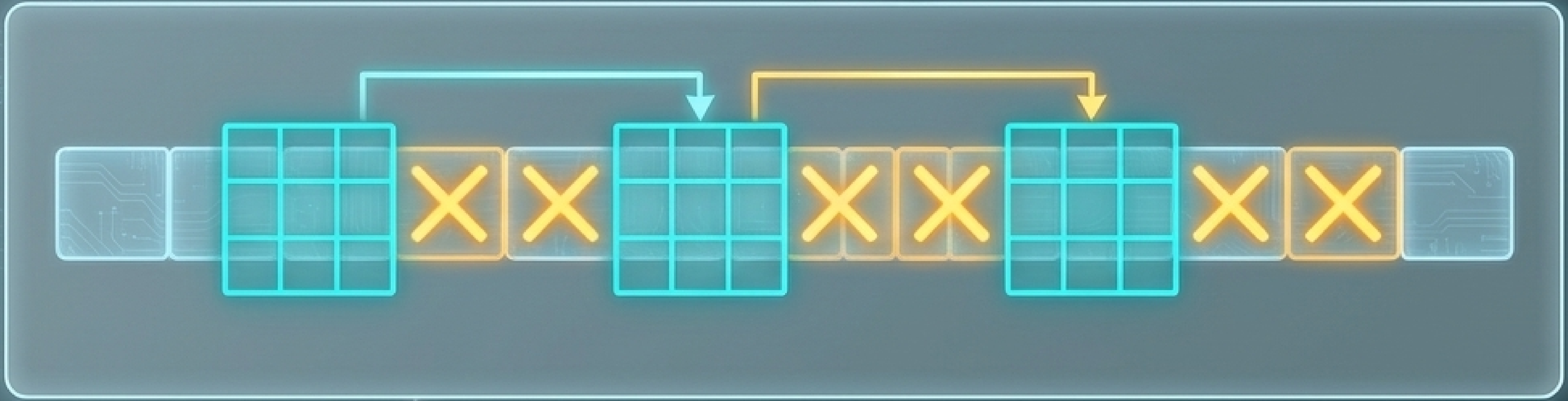
`By setting p = floor(k/2), the spatial output perfectly matches the input resolution.`

Adjusting the Focal Step: Subsampling with Stride (s)

Unit Strides (s = 1)



Strided Subsampling (s = 2)



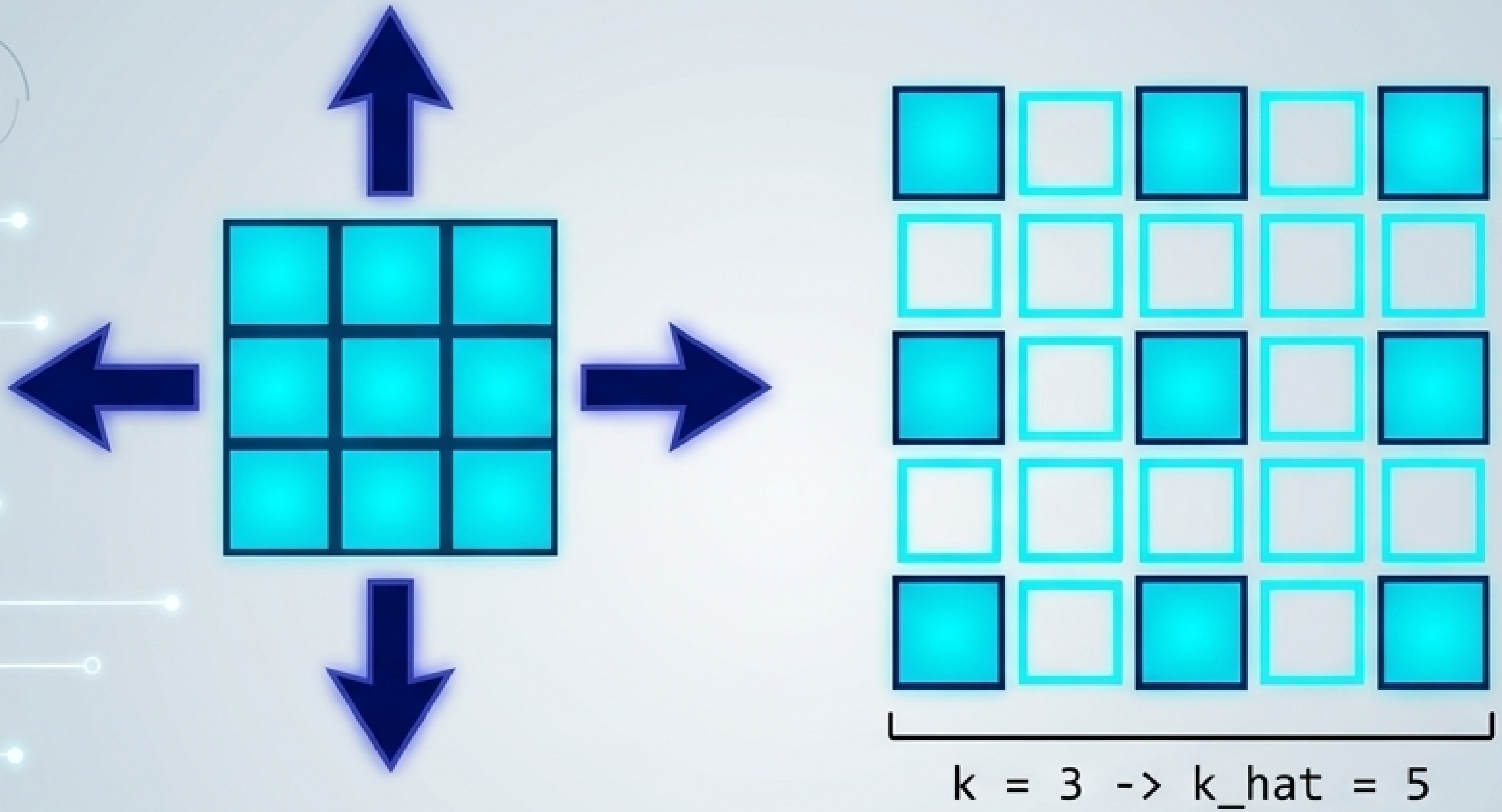
Efficiency

Strides (s > 1) constitute a form of spatial subsampling. Instead of translating by increments of 1, the kernel hops.

The Math

By dividing the spatial plane by s, the output dimensions shrink exponentially, reducing memory footprint and compute load deeper in the network.

The Net Expansion: Multi-Scale Context via Dilation (d)



The Atrous Inflation

Dilated convolutions inflate the kernel by inserting (d-1) spaces between kernel elements.

The Equation

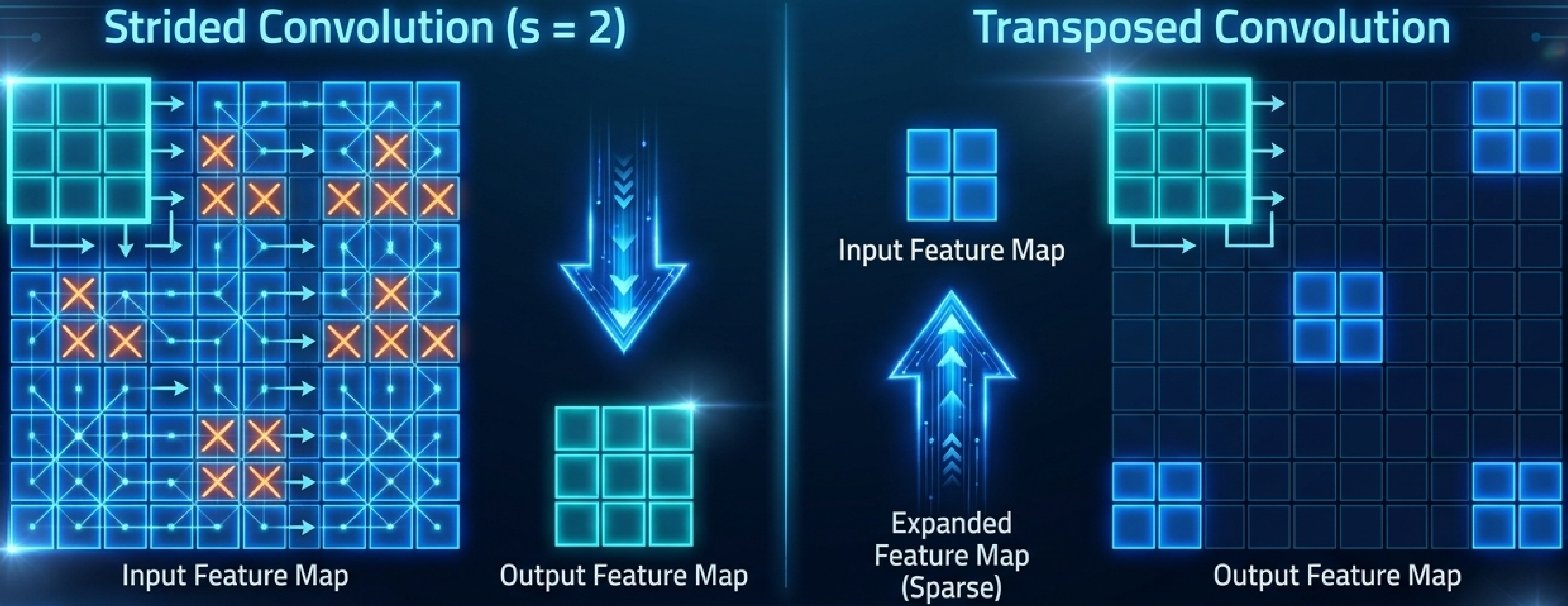
The effective footprint of the kernel becomes `k_hat = k + (k-1)(d-1)`.

The Advantage

Cheaply increases the receptive field without adding computational parameters, preventing the loss of resolution associated with pooling.

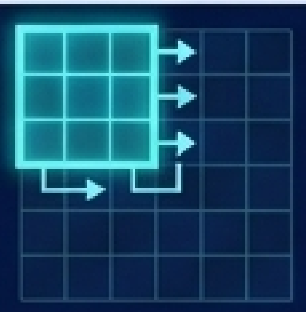
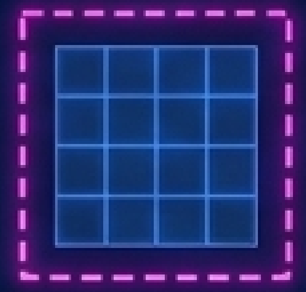
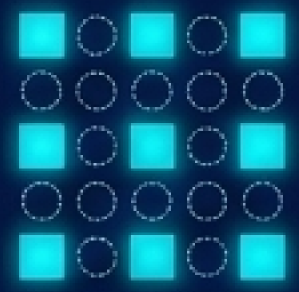
Light Optic Workbench

Reversing the Lens: Fractional Striding (Transposed)

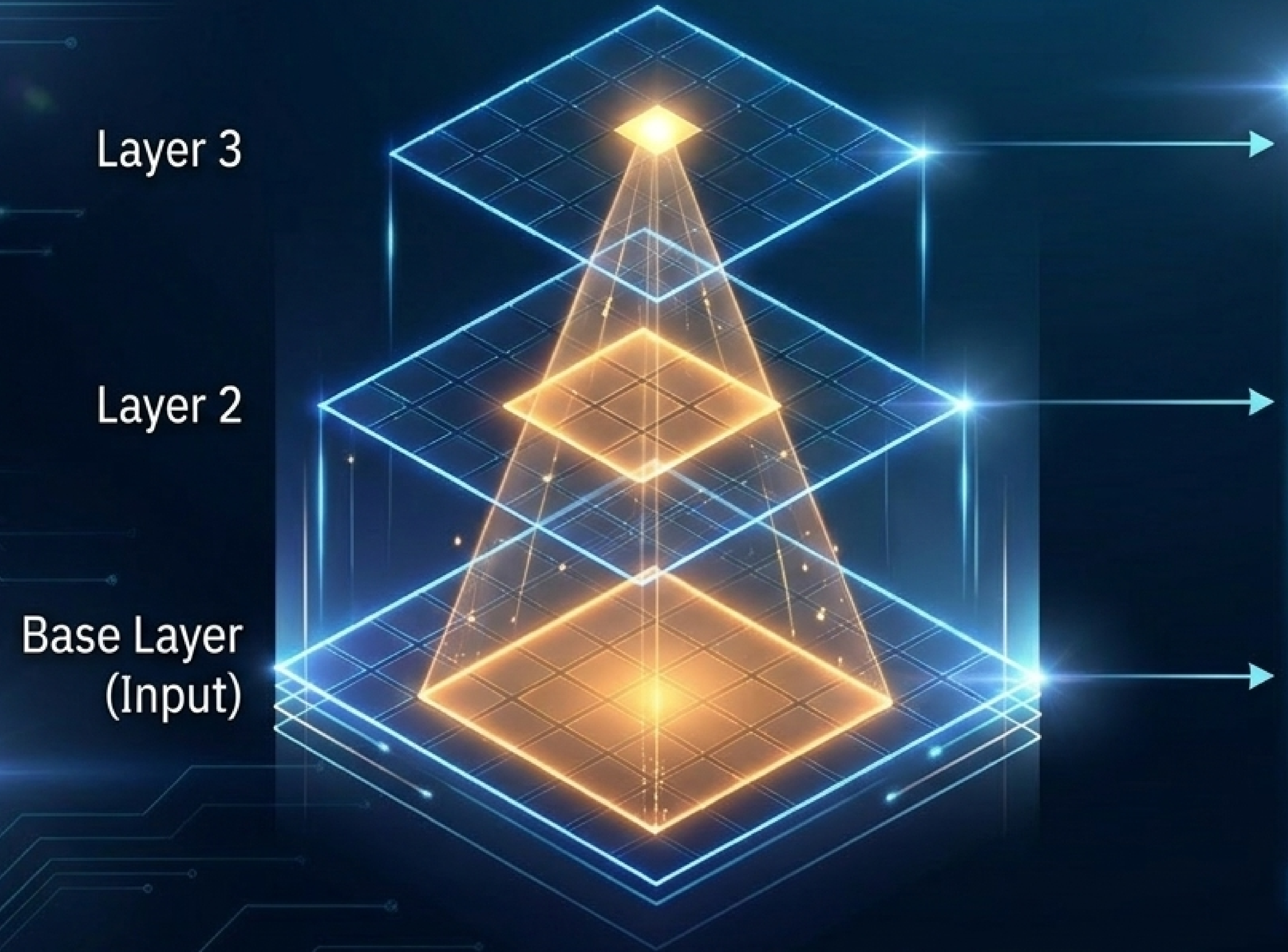


- Transposed convolutions map from a low-dimensional space back to a high-dimensional space (e.g., in autoencoders or image segmentation).
- Instead of striding over pixels to shrink the output, we simulate fractional strides ($s < 1$) by padding the input with zeros between the active pixels, then convolving normally.

The Spatial Operations Toolbelt

Visual Icon	Operation & Purpose	Effect on Output Size	Example Use Case
	Standard Convolution: Local feature extraction.	Slight reduction (i - k + 1)	Early layer edge detection.
	Padded Convolution: Preserve spatial resolution.	Maintains exact size (o = i)	Deep ResNet intermediate blocks.
	Strided Convolution: Spatial downsampling & compression.	Reduces size by a factor of s	Replacing Max Pooling layers.
	Dilated Convolution: Wide context aggregation.	Expands receptive field without losing resolution.	Semantic Segmentation (WaveNet).

From Arithmetic to Perception: The Theoretical Receptive Field



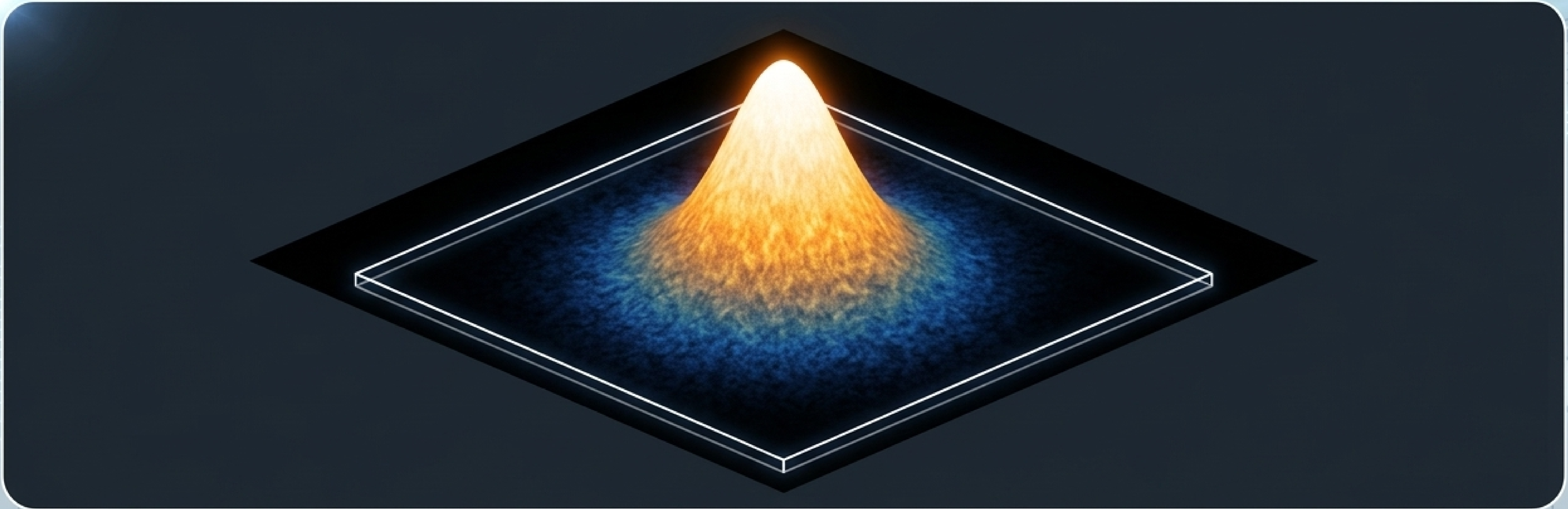
The Definition

The Theoretical Receptive Field (TRF) is the absolute mathematical region in the original input image that can potentially influence a single output unit deep in the network.

Linear Growth

Stacking multiple layers increases the TRF size linearly (O(n)). For every layer with kernel k, the TRF boundary grows by exactly k-1.

The Gaussian Revelation: The Effective Receptive Field (ERF)

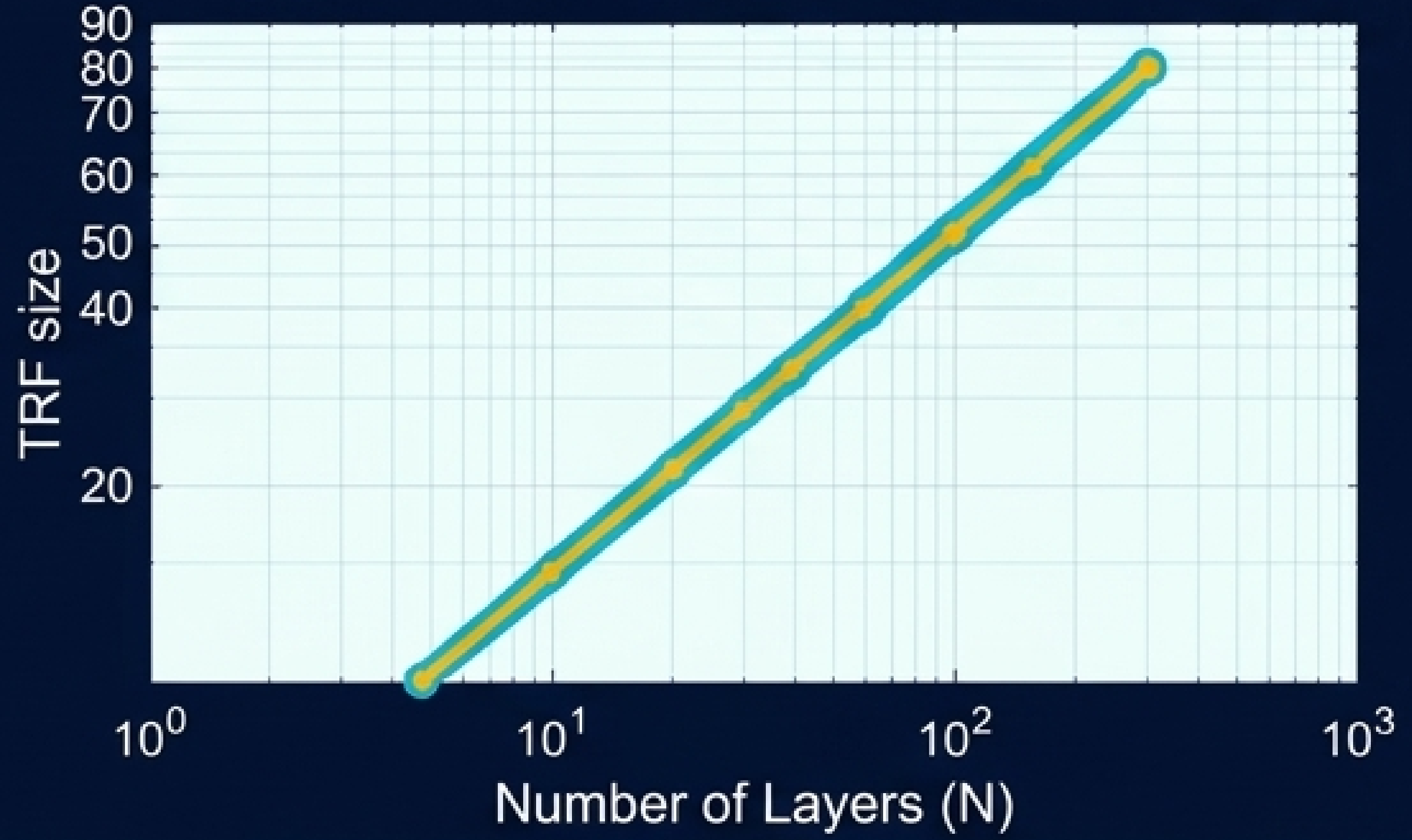


The Illusion of Bounds	Multiplying Paths	The Result
While the Theoretical Receptive Field outlines a hard geometric boundary, the Effective Receptive Field operates as a Gaussian distribution.	Central pixels propagate information to the output through exponentially more forward/backward paths than pixels situated on the outer edges.	A CNN only truly focuses on a fraction of its theoretical field of view. Outer edge pixels have a near-zero gradient impact on the final output.

Analyzing the Impact: TRF vs. ERF

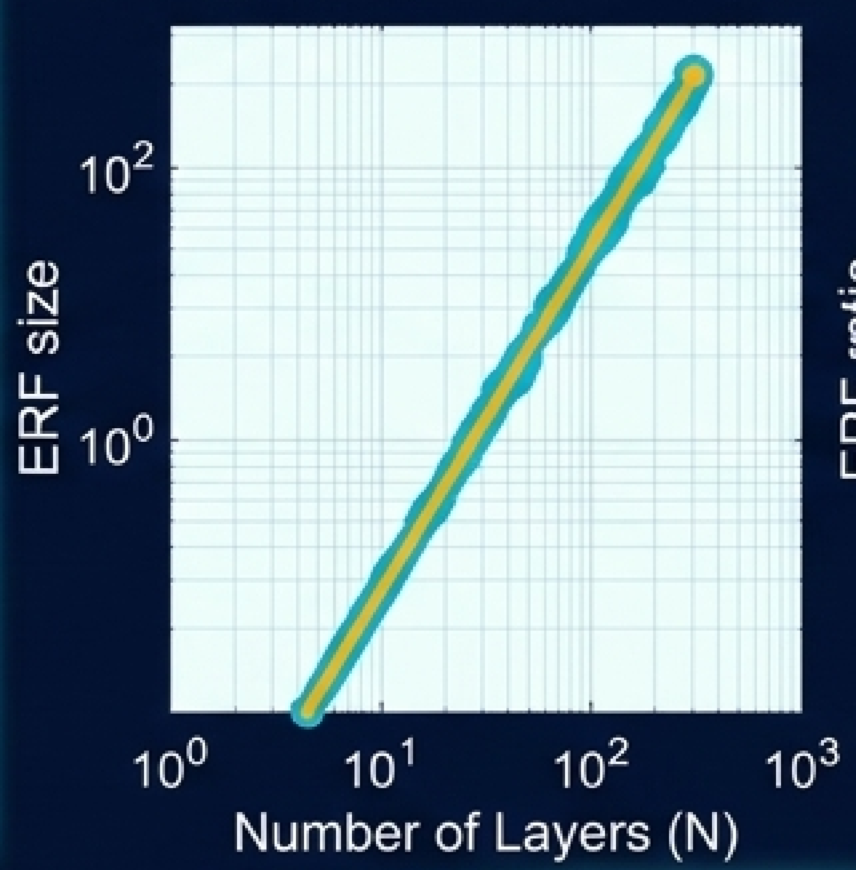
Theoretical Receptive Field (TRF)

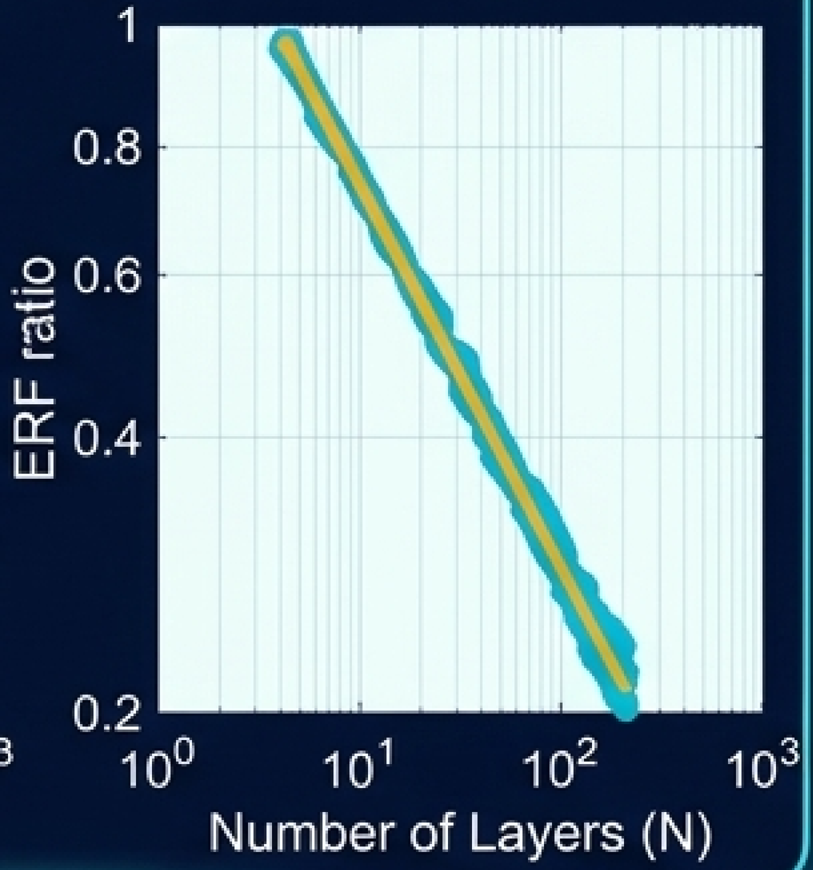
- **Shape:** Square Grid.
- **Growth Rate:** $O(n)$ — Grows linearly with the number of layers.
- **Implication:** What the architecture suggests. Assumes all pixels within the bounding box contribute equally to the final prediction.



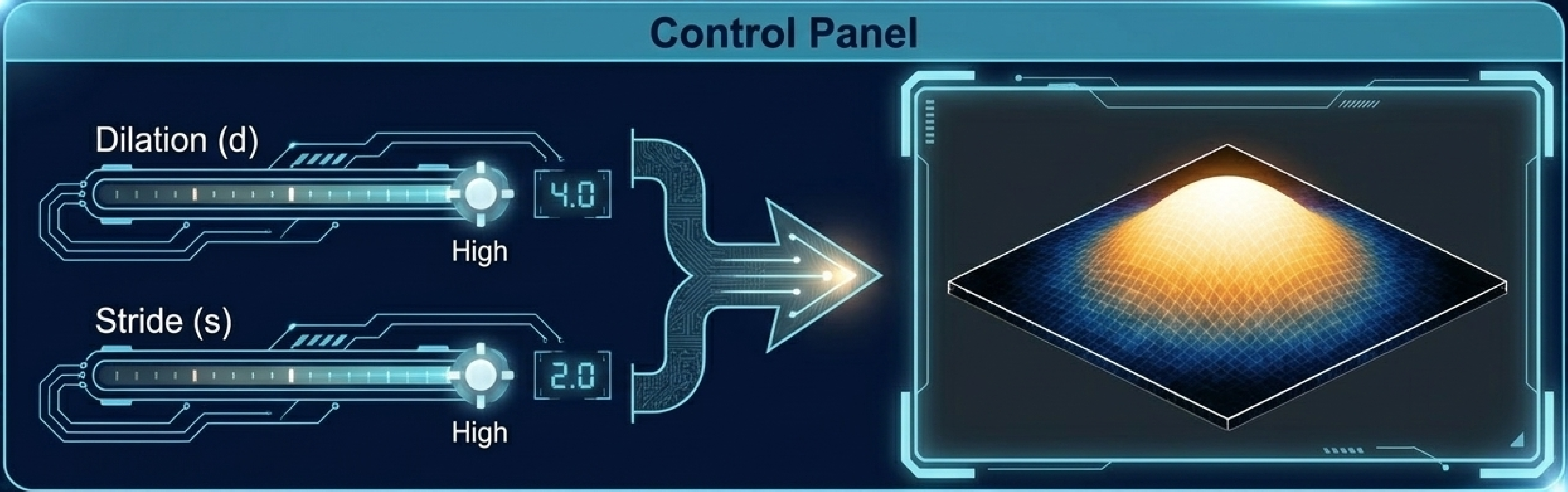
Effective Receptive Field (ERF)

- **Shape:** Gaussian Heatmap.
- **Growth Rate:** $O(sqrt(n))$ — Grows at the square root of layers.
- **Implication:** What the network actually uses. The relative ratio of ERF to TRF surprisingly shrinks as the network gets deeper.



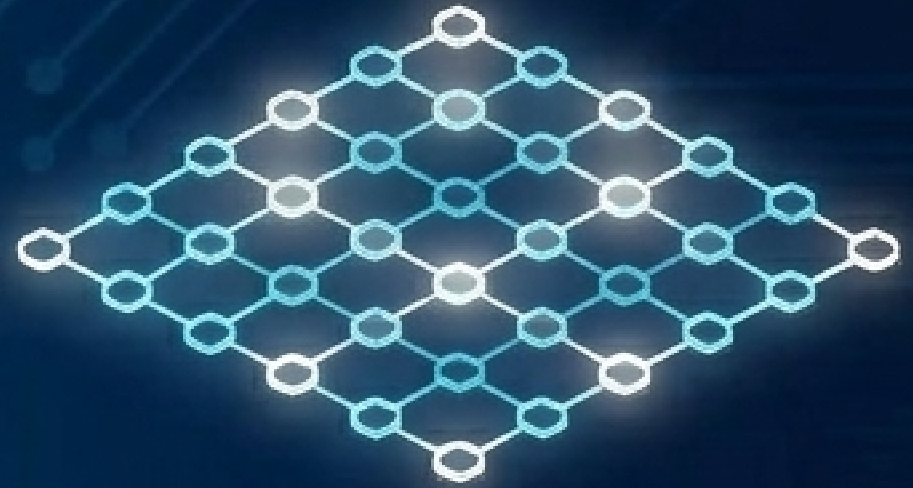


Tuning the Network's Eye



- **The Challenge:** Dense prediction tasks require large, robust receptive fields, but the ERF's Gaussian decay means the network naturally suffers from 'tunnel vision'.
- **The Steering Mechanism:** Spatial arithmetic is our direct control surface to counteract this decay.
- **The Solution:** Integrating Strides and Dilated Convolutions are the most potent architectural methods to artificially force the Effective Receptive Field to rapidly expand, pushing critical context back into the network's focal center.

The Evolution of Artificial Vision



AlexNet

Heavy striding and aggressive spatial reduction. Overlapping pooling (s=2, z=3).

ResNet

Deep stacks of 3x3 padded convolutions to slowly build linear TRF.

Modern Dilated Nets

Utilizing d=2, 4, 8 to capture massive, high-resolution context without losing spatial fidelity.

Spatial arithmetic is not arbitrary math; it is the physical engineering of an artificial lens. By manipulating kernels, padding, strides, and dilations, we dictate not just the size of the feature maps, but the very nature of how a neural network perceives its world.